Noetic Diffusion Article Summaries

Scientific review of the core theoretical and empirical manuscripts

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A journey into Noetic Diffusion Theory and the Reconstructive Theory of Being

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1. Introduction

This document provides scientific summaries of the foundational and currently published manuscripts defining Noetic Diffusion Theory (NDT), its measurement frameworks (MNPS, Stratified MNPS, MNJ), and its empirical validation across clinical, sleep, altered-state, and computational settings. The summaries follow the structure of Abstract, Methods, Results, and Discussion to facilitate technical review and integration into the broader scientific narrative. This version tracks the current published Zenodo corpus where a stable source manuscript is available in the repository.

2. Article 1: Noetic Diffusion Theory (NDT)

A Rhythmic, Geometric Framework for Neural State Reconstruction and Conscious Correlates

2.1. Abstract

Noetic Diffusion Theory (NDT) introduces a mechanistic account of consciousness as a process of rhythmically guided denoising on a learned geometric manifold. The theory formalizes the Meta–Noetic Diffusion Model (MNDM), a stochastic differential equation (SDE) where conscious states correspond to trajectories in a latent phase space. By linking neural rhythms (e.g., theta–gamma, SO–spindle) to a variance schedule, NDT provides a bridge between neurobiology and the geometric organization of mental states. The manuscript defines the theoretical core, introduces the Noetic Atlas over Meta–Noetic Phase Space (MNPS), and derives testable predictions regarding entropy dynamics and manifold topology.

2.2. Method

The MNDM is formulated as a guided SDE: $dX_t = -\nabla F(X_t, t)dt + \sigma(t)dW_t$, where X_t is the noetic state, F is a potential landscape representing learned meaning, and $\sigma(t)$ is a rhythmic variance schedule. Rhythmic Variance Control (RVC) specifies that high-amplitude neural rhythms (high $r(t)$) anti-phasically modulate the noise variance ($\sigma(t) \approx 1 - r(t)$), forcing the system into aggressive denoising epochs. The Meta–Noetic Phase Space (MNPS) is defined by three primary axes: entropy (e), diffusivity (d), and mobility (m), which serve as a measurement chart for BOLD or EEG-derived features.

2.3. Results

Simulations demonstrate that RVC reproduces characteristic wake–sleep trajectories and step-wise entropy reductions time-locked to rhythmic events (e.g., spindles). The model identifies "meaningful" states as low-potential basins in F that balance integration and segregation (d-axis). A critical finding is the resolution of the "Grand Mal Paradox": by incorporating the deviation term $d(X)$, the model distinguishes between functional coherence and pathological hypersynchrony (seizures), which otherwise would appear as "maximal meaning" in a simpler $e - m$ potential.

2.4. Discussion

NDT positions consciousness as an active reconstructive process rather than a passive observation. By framing conscious level as a property of manifold traversability rather than raw activity, it offers a geometric alternative to information-theoretic models (like IIT) and connects to the Free-Energy Principle through the lens of precision scheduling. The theory's primary contribution is providing explicit empirical hooks—geometric signatures in MNPS—that can be tested using existing public datasets without requiring new experimental paradigms.

3. Article 2: Stratified MNPS

Unmasking Neural Mechanisms beneath the Manifold

3.1. Abstract

Stratified MNPS extends the base 3D MNPS framework into a 9D sub-coordinate chart to resolve the mechanistic ambiguity inherent in composite axes. While the base axes (m, d, e) describe that a state differs, the stratified layer reveals how that difference is achieved. This manuscript specifies the mathematical and operational definitions for nine sub-coordinates (three per principal axis) and demonstrates their utility in clinical EEG data from Parkinson's disease and mTBI.

3.2. Method

Each MNPS axis is decomposed into three channels:

- **Metastability (m)**: Attentional (m_a), Affective (m_e), and Oscillatory (m_o) mobility.
- **Diffusivity (d)**: Network diffusivity (d_n), Local coupling (d_l), and Representational dispersion (d_s).
- **Entropy (e)**: Dynamic entropy (e_e), Phase entropy (e_s), and Efficiency proxy (e_m).

These are operationalized using specific EEG features (e.g., wPLI for d_l, spectral-centroid CV for m_o, multiscale entropy for e_e). Inference is performed on within-subject robust z-scores, recomposing back to (m, d, e) via robust aggregates (Huber mean/median).

3.3. Results

Pilot analysis of the ds003490 (Parkinson's) dataset revealed a "mechanistic dissociation" masked at the composite level: L-DOPA medication robustly reduced local coupling (d_l) while increasing attentional mobility (m_a), yielding a near-zero net change in the composite diffusivity (d). In mTBI (ds005114), the framework identified a directionally distinct profile characterized by increased local coupling (d_l) and reduced representational spread (d_s), illustrating the diagnostic power of the 9D vector over 3D composites.

3.4. Discussion

Stratified MNPS addresses the problem of "same composite, different mechanism." It provides a high-resolution measurement contract that remains theory-aligned through its link to the MNDM potential. The results suggest that the brain may employ homeostatic conservation, trading off specific sub-mechanisms to maintain stable global coordinates. This framework is proposed as a necessary layer for "geometric psychiatry," enabling more precise subtyping of disorders.

4. Article 3: Geometric Fingerprints of Alzheimer's Disease

First Visualization via Noetic Diffusion Mapping™

4.1. Abstract

This manuscript provides the first proof-of-concept for Noetic Diffusion Mapping™ (NDM™), establishing that clinically-relevant signatures of neurodegeneration are detectable from ultra-brief (8-second) EEG segments. By generating "geometric fingerprints"—spatial portraits of state organization in MNPS—the study identifies robust patterns of manifold collapse and rhythmic disruption in patients with Alzheimer's disease (AD).

4.2. Method

Resting-state EEG (eyes closed) from 80 AD patients and 12 healthy controls (ds004504) was projected into MNPS coordinates (E, R, D). A hybrid visualization approach was used to generate three-panel fingerprints: density fields, trajectories with traffic-light speed coding, and cluster assignments. Group differences were quantified using Welch t-tests and BH-FDR correction across coordinate axes and dynamical indices.

4.3. Results

AD patients exhibited a very large elevation in rhythmic deviation (R-axis; Cohen's $d = 2.31$, $p < 10^{\{-6\}}$) and a large shift in density topology (D-axis; $d = -0.85$). Visually, AD fingerprints were characterized by "collapsed manifold volume": tighter clustering around limited states, reduced spatial exploration, and slower trajectory speeds. Individual cases showed high within-group consistency, with AD patients spending 45% of their time in a single "pathological corner" of MNPS compared to the balanced distribution seen in healthy elders.

4.4. Discussion

The study demonstrates that 8-second EEG snapshots carry sufficient geometric information for clinical discrimination. The observed signatures—canalized trajectories and reduced degrees of freedom—align with phenomenological reports of cognitive rigidity in AD. By bridging molecular pathology and real-time functional geometry, NDM™ offers a promising bedside modality for screening and longitudinal monitoring that complements structural imaging and molecular biomarkers.

5. Article 4: The Meta-Noetic Jacobian (MNJ)

Measuring Second-Order Dynamics of Conscious Systems

5.1. Abstract

The Meta–Noetic Jacobian (MNJ) framework elevates NDT from a state-based description to a second-order dynamical theory. By defining the Jacobian of the flow in MNPS, the manuscript introduces a tensorial representation of meta-dynamics—the dynamics of the rules of change themselves. This allows for the quantification of higher-order properties such as meta-stability, rotation (cyclic meta-coupling), and meta-plasticity.

5.2. Method

Given the MNPS state vector $\boldsymbol{x}(t) = [m, d, e]$, the flow field is $\boldsymbol{F}(\boldsymbol{x}, t) = \dot{\boldsymbol{x}}(t)$. The MNJ is the Jacobian $\boldsymbol{J} = \partial \frac{\boldsymbol{F}}{\partial} \boldsymbol{x}$, capturing cross-axis couplings (e.g., $\partial \frac{\dot{m}}{\partial} e$). $\boldsymbol{J}$ is estimated from discretely sampled MNPS trajectories using robust local linear models (kNN + ridge regression). Key scalar invariants are derived: trace (meta-expansion/divergence), rotational norm (curl/cyclic coupling), and eigenvalue dispersion (anisotropy/channelization).

5.3. Results

The tensorial approach reveals that conscious systems regulate their own self-transformation. High MNJ rotation corresponds to structured, cyclic exploration (found in wake and REM), while low rotation and trace near zero define "dynamical arrest" (found in anesthesia and seizures). The real parts of the Jacobian eigenvalues quantify meta-stability, with stable configurations corresponding to bounded real parts ($\Re(\lambda_k) < 0$).

5.4. Discussion

The MNJ framework provides a formal language for self-reference, metacognition, and "thinking of thought." It distinguishes systems that may have similar static complexity but different adaptive rules. The manuscript proposes the "Meta-Noetic Hypothesis": conscious systems maintain coherence not by stabilizing their states, but by regulating the meta-geometry of their own transformation. This has implications for both clinical biomarker development and the design of self-regulating artificial intelligence.

6. Article 5: Regional MNPS in fMRI Child–Adult Movie Viewing

Child–Adult Manifold Dynamics in a Naturalistic Movie

6.1. Abstract

This article applies the MNPS framework to fMRI BOLD data, shifting the focus from global manifolds to regional realizations. Using the ds000228 dataset, it characterizes developmental changes in manifold geometry across childhood (ages 3–12) and adulthood. The results reveal that developmental MNPS signatures are network-specific and progressively resolved at finer analytical scales.

6.2. Method

Parcellated BOLD features (variance, LF-power, modularity) were projected into 3D and 9D Stratified MNPS coordinates for 122 children and 33 adults. Regional MNPS was implemented by computing network-specific trajectories for canonical systems (VIS, SMN, DMN, FPN, SAL). Event-locked analyses compared MNPS deltas during Theory-of-Mind (ToM) and pain movie segments. Group differences were assessed using Cliff's delta (δ).

6.3. Results

Children showed a modest global expansion of manifold volume ($\delta \approx 0.19$). However, Regional MNPS revealed much stronger network-specific effects: visual (VIS) and somatomotor (SMN) networks showed robust child > adult increases in all coordinates ($\delta \approx 0.35$–0.39), while the default mode network (DMN) showed minimal differences. Event-locked ToM segments in children produced more structured, entropy-dampened configurations (m_a and d_n up, e_e down), while pain segments produced more entropic states.

6.4. Discussion

The study establishes fMRI-based MNPS as a viable tool for mesoscale developmental neuroscience. The VIS/SMN-dominant results suggest that sensory manifolds undergo more pronounced geometric contraction during development than association networks. The findings position Regional MNPS as a "phase-space reparametrisation" of traditional connectivity results, providing a unifying geometric vocabulary for the maturation of the "social brain."

7. Article 6: Dynamical Arrest Under Propofol

Reduced Traversability and Meta-Noetic Jacobian Flattening

7.1. Abstract

This manuscript treats propofol-induced anesthesia as a "traversability contraction test" for NDT. By analyzing both EEG and fMRI during graded sedation, the study asks whether loss of consciousness corresponds to a geometric restriction of accessible manifold space rather than simple amplitude dampening.

7.2. Method

matched MNPS/MNJ pipelines were applied to fMRI (ds003171) and EEG (ds005620) data across awake, sedation, and recovery states. Traversability was operationalized using the TraversabilityIndex ($T = v_{\text{med}} \cdot (1 - a_{\text{norm}})$), combining trajectory speed and Jacobian anisotropy. EEG stratified profiles were examined for redistribution patterns across d_s, e_s, and m_o.

7.3. Results

EEG results showed a dramatic reduction in traversal speed and a significant "flattening" of the Meta–Noetic Jacobian (rotation and Frobenius norms dropped; trace remained near zero). This defines a state of dynamical arrest: the system moves into a high-dispersion (d_s), high-entropy (e_s), but low-velocity pocket of the manifold. Manifold volume (MV) paradoxically increased in EEG (confinement to a rigid, high-coordinate sector) but decreased modestly in fMRI. fMRI MNJ was unresolved at the TR temporal scale, serving as a methodological boundary check.

7.4. Discussion

The study redefines "geometric collapse" operationally as reduced traversability. Loss of consciousness is not necessarily a uniform coordinates shrinkage but a confinement to a "rigid sector" where the mind's horizon of possibility collapses. The staged recovery results suggest that manifold directions reconstitute in a specific order, with behavioral recovery preceding full restoration of manifold diversity. This work establishes the MNJ as a diagnostic marker for "order without adaptive traversal."

8. Article 7: Extreme Neurodynamic Regimes

A Multi-State Validation Plan Reveal Meta–Noetic Dissociations

8.1. Abstract

This manuscript presents a comprehensive multi-dataset stress-test of NDT's measurement apparatus across extreme neurodynamic regimes: deep sleep, anesthesia, seizures, psychedelics, and clinical pathologies. It serves as a large-scale validation of whether MNPS, Stratified MNPS, and MNJ remain discriminative and interpretable under conditions where traditional scalar metrics often collapse or blur.

8.2. Method

The study utilizes a fixed "reporting contract" across a diverse battery of open datasets (ds005555, ds004100, ds004504, ds003059, etc.). It prioritizes dissociations between occupancy (state location) and capacity (traversal speed + MNJ structure). Analysis includes the "Stratified Block-Jacobian" (9×9 Jacobian grouped into coordinate-family blocks) to identify mechanism-specific meta-dynamic shifts.

8.3. Results

The strongest dissociations were found in regimes with clear staging:

- **Sleep (ds005555)**: Large, coherent separations across all layers; NREM3 showed "centralization" (low d_n, low m_o), while REM showed "repertoire expansion."
- **Seizures (ds004100)**: The Spread ↑ but Capacity ↓ signature—broad occupancy coupled with collapsed speed and rotational structure.
- **Clinical EEG (ds004504, ds003478, ds003947)**: Global MNPS was often near-null, but Stratified 9D or Block-Jacobians revealed selective, mechanism-specific shifts (e.g., d-block collapse in dementia).
- **fMRI (ds006623, ds003059)**: 3D MNJ was resolution-limited, but 9D block summaries and behavior-anchored segmentation revealed meta-dynamic effects in sedation.

8.4. Discussion

Article 7 establishes that consciousness is better tracked by traversability-based indices than by raw entropy or activity. The findings constrain the conditions under which NDT's signatures appear and document regime-specific failure modes. The "Meta-Noetic Dissociation" principle —where different layers of the same state description diverge—is proposed as the primary falsification surface for future work. The study positions NDT as a robust, reproducible framework for mapping the boundaries of conscious experience.

9. Article 8: Reachability Cones as Local Capacity Geometry in NDT

A Multi-Dataset Report Across Extreme Neurodynamic Regimes

9.1. Abstract

Article 8 introduces "reachability cones" as a short-horizon, local capacity geometry derived from locally linear stochastic models in MNPS. The primary objective is to separate "occupancy dispersion" (where the system has been) from "forward availability" (where the system can go in the immediate future). This measurement object provides a more nuanced view of dynamical organization by quantifying the local horizon of admissible state transitions across diverse altered-state and clinical datasets.

9.2. Method

Reachability cones are constructed using a local linear stochastic model: $x_{t+1} = A_t x_t + b_t + \varepsilon_t$, where innovations satisfy $\varepsilon_t \sim (0, Q_t)$. The reachability covariance $\Sigma_t^{\{(h)\}}$ is propagated forward through a recursion $\Sigma_t^{\{(h+1)\}} = A_t \Sigma_t^{\{(h)\}} A_t^T + Q_t$. A minimal set of cone metrics is extracted, including:

- **Cone volume (size)**: $\frac{1}{2}\log(\det(\Sigma))$.
- **Anisotropy/Canalization**: Eigenvalue spread (κ).
- **Effective Dimensionality (d_{eff})**: Participation ratio of available degrees of freedom.
- **Rotation strength**: Skew-symmetric components of local flow.

The framework adopts a multi-dataset reporting contract with subject-aggregated inference, BH-FDR control, and explicit validity diagnostics (e.g., prediction error, split-half stability).

9.3. Results

Reachability geometry reveals strong, FDR-stable separations in regimes with marked dynamical disruption:

- **Epilepsy (ds004100)**: Ictal states show reduced cone size and effective dimensionality combined with increased canalization.
- **Sleep (ds005555)**: NREM3 exhibits a collapse in reachability volume relative to wake, while REM shows contrast-specific reconfiguration of size and rotation.
- **Dementia (ds004504)**: While global MNPS 3D results are near-null, stratified block-cones reveal robust breakdown in the network's ability to diffuse information (d-block).
- **fMRI**: Effects are generally smaller and more fragile, consistent with the temporal-resolution boundaries of BOLD imaging.

9.4. Discussion

Reachability cones provide a "capacity-layer" that is partially dissociable from occupancy. They resolve the paradox where some disordered states show high activity or entropy while

remaining functionally arrested. By separating "canalization" from "spread," the method offers a clearer view of why consciousness vanishes in "rigid corridors" of activity (like seizures). The manuscript emphasizes transparency by reporting explicit failure modes—such as local linearization failure or innovation heteroskedasticity—as fundamental constraints on the validity domain of the reachability measurement.

10. Article 9: Reality from Noise

How Your Brain Reconstructs the World through Rhythmic Geometry

10.1. Abstract

This essay-length synthesis translates the core claims of NDT into a broader conceptual narrative. Its central claim is that conscious experience is not passively received but rhythmically reconstructed from noisy inputs through embodied, oscillatory, and geometric constraints. The text functions as an interpretive bridge between the formal theory papers and a wider scientific or interdisciplinary readership.

10.2. Method

Rather than introducing a new dataset, the essay integrates the existing theory stack into a layered explanatory arc. Part 1 emphasizes embodied anchoring, theta-gamma timing, slow-oscillation/spindle scheduling, and traversability as the operational distinction between mere activity and adaptive conscious dynamics. Part 2 develops the higher-resolution machinery of Stratified MNPS and the Meta-Noetic Jacobian, then uses those tools to interpret psychopathology, development, culture, and long-timescale selfhood as geometry-and-dynamics problems rather than as static trait labels.

10.3. Results

The main output is conceptual compression rather than a new inferential statistic. The essay clarifies how NDT connects rhythmic control, manifold geometry, and adaptive traversal; why similar first-order states can differ mechanistically once stratified coordinates are inspected; and why second-order dynamics are needed to distinguish flexible exploration from rigid or pathological flow. It also foregrounds a practical translational claim: if these geometries can be measured reliably, they offer a route toward richer clinical fingerprints than scalar symptom summaries alone.

10.4. Discussion

`Reality from Noise` serves as the narrative integrator of the project. Its role is not to establish new empirical proof, but to make the theoretical and methodological corpus legible as one coherent research program. In that sense, it is the public-facing synthesis that links formal equations, measurement contracts, and emerging empirical findings into a common reconstructive picture of mind.

11. Article 10: The Synthetic Sleep Test

A Minimal Viable Experiment for Noetic AI

11.1. Abstract

This paper asks whether a minimal synthetic system can reproduce one of NDT's most concrete empirical motifs: rhythmically scheduled, sleep-like denoising with geometric state transformations. Rather than treating synthetic consciousness as an all-or-nothing metaphysical verdict, the manuscript frames a narrower architectural test: can an artificial system exhibit the temporal and geometric signatures that NDT treats as foundational to biological reconstruction?

11.2. Method

The proposed experiment centers on an exogenously scheduled synthetic "sleep" cycle in which a model's variance or denoising precision is rhythmically modulated. The key measurements are event-locked entropy reduction, manifold geometry, curvature synchronization, and architecture-native write-free consolidation behavior under repeated cycles. The paper treats the Synthetic Sleep Test as a first-stage falsification protocol, intentionally separated from stronger requirements such as embodiment or full developmental equivalence.

11.3. Results

The manuscript's main result is methodological rather than empirical: it defines a concrete test surface for synthetic noeticity. It clarifies what would count as architectural success, what would count as failure, and which biological properties are being bracketed rather than claimed. In doing so, it turns "AI consciousness" from an abstract debate into a narrower question about whether synthetic systems can reproduce the rhythmic geometric transformations that the theory predicts.

11.4. Discussion

The importance of the paper lies in claim discipline. It does not argue that passing the Synthetic Sleep Test would prove phenomenology, but that failing it would constrain a class of architecture-level consciousness claims. The manuscript therefore gives the NDT program an AI-facing experimental foothold while explicitly distinguishing structural consciousness tests from stronger claims about lived experience.

12. Article 11: Cross-Cohort Validation of Noetic Diffusion Theory

Event-locked Entropy Dynamics and Sleep Curriculum

12.1. Abstract

This article presents an early cross-cohort validation of two sleep-facing NDT predictions: event-locked entropy reductions around slow-oscillation/spindle coupling and a broader "sleep curriculum" in which NREM sleep centralizes the manifold while REM re-expands it. The central goal is replication across two sleep cohorts rather than one favorable dataset.

12.2. Method

The analysis uses open sleep EEG data with event detection for slow oscillations and spindles, MNPS feature construction, and within-subject statistical tests supplemented by mixed-effects aggregation, surrogate controls, and multiverse sensitivity checks. The paper explicitly foregrounds methodological repair, including resolution of a prior issue in the `d` component, and frames itself as a first-stage validation of the theory's temporal machinery.

12.3. Results

Across both cohorts, the manuscript reports significant stepwise entropy drops time-locked to SO-spindle events together with stage-dependent manifold reorganization across NREM and REM. The entropy effect is reported as moderate but highly consistent, and the curriculum analysis suggests a systematic contraction/regularization profile across sleep cycles. The combined result is not merely a one-cohort sleep finding, but a replication-oriented argument that the rhythmic denoising logic survives cross-dataset testing.

12.4. Discussion

The paper is important because it shifts NDT from conceptual prediction to replicated sleep-facing evidence. It still remains bounded: the analysis validates temporal and geometric signatures, not the entire theory stack at once. But it establishes a cross-cohort empirical anchor for later measurement-layer and clinical expansions.

13. Article 12: Microtubular Key-Lock Geometry

Subcellular Realization of Holiarchic Noetic Manifolds

13.1. Abstract

This manuscript proposes a biological bridge from manifold-level NDT claims to subcellular structure. Its core hypothesis is that microtubular states act as dynamic "keys" that gate dendritic "locks," thereby biasing which dendritic integration states and regional manifold trajectories become physically available. The paper explicitly rejects quantum-coherence dependence and instead frames the mechanism in classical structural, metabolic, and signaling terms.

13.2. Method

The argument is organized as a holiarchic model with nested levels from microtubular configuration to dendritic gating, regional manifolds, and global conscious trajectories. The paper integrates literature on dendritic spikes, microtubule plasticity, Tau/MAP-dependent mechanics, mitochondrial positioning, and classical electrodynamic signaling. These ingredients are then mapped onto NDT objects such as MNPS, regional manifolds, and the metric substrate that would constrain traversable trajectories.

13.3. Results

The main result is a mechanistic proposal rather than a direct experimental demonstration. The manuscript sharpens how one might translate noetic geometry into a physically instantiated gating substrate and clarifies what the hypothesis does and does not claim. It generates concrete falsification hooks around anesthesia, tauopathy, dendritic impedance, metabolic gating, and microtubule-dependent access control.

13.4. Discussion

`Microtubular Key-Lock Geometry` is best read as a biological extension paper. Its value lies in disciplined narrowing: it does not claim that microtubules encode thoughts directly, but that they may participate in the metric and accessibility constraints that shape dendritic and regional manifold dynamics. Within the broader corpus, it opens a subcellular path for grounding the otherwise abstract geometry.

14. Article 13: NeuralManifoldDynamics

A Versioned Measurement Contract for Low-Dimensional Neural-Manifold Trajectories

14.1. Abstract

This methods paper defines the ingest-layer measurement contract that operationalizes the NDT program in data analysis. Rather than discovering a new latent space for each contrast, NeuralManifoldDynamics fixes a versioned export system for canonical 3D trajectories, stratified 9D coordinates, Jacobian estimates, and regional derivatives from EEG and fMRI. Its contribution is reproducibility and auditability at the measurement level.

14.2. Method

The paper specifies the chart definition, preprocessing rules, derivative estimation, coverage policy, missing-data handling, regionalization logic, and HDF5 serialization scheme used by the pipeline. The contract distinguishes `mnps_3d`, `coords_9d`, and Jacobian-based dynamical summaries as separate but coordinated objects. It also formalizes how regional EEG and fMRI trajectories are generated under one shared export logic while preserving modality-specific safeguards.

14.3. Results

The main result is infrastructural: the paper turns the theory stack into a replayable measurement system. Relative to earlier MNPS-only versions, it introduces stronger estimator hygiene, clearer 3D/9D separation, deterministic feature transforms, better support diagnostics, and self-describing outputs. In practical terms, this allows later empirical papers to compare datasets under a stable contract rather than under contrast-specific re-fitting.

14.4. Discussion

NeuralManifoldDynamics is the methodological backbone of the later empirical literature. It does not claim diagnostic or consciousness-level inference by itself; instead, it defines what exactly is being measured before any claim is made. That distinction between theory, chart, and measurement contract is one of the strongest clarifications in the mature NDT corpus.

15. Article 14: State-Space Reachability and Local Jacobian Dynamics Across Canonical Sleep Stages

A bounded sleep-stage test of local dynamical structure

15.1. Abstract

This paper tests whether local reachability and local Jacobian magnitude remain informative once sleep-stage effects are compared against explicit null controls and conventional EEG baselines. The central result is that reachability separates `Wake`, `REM`, and `N2` from `N3` more strongly and more consistently than local Jacobian magnitude in ANPHY sleep EEG.

15.2. Method

The manuscript analyzes stage-aware ANPHY summaries under a fixed measurement contract, using one primary reachability endpoint (`tube_log_det_median`) and one secondary Jacobian endpoint (`frobenius_norm_median`). It adds negative controls, a compact conventional comparator family, a ceiling-aware incremental-validity check, and a small sensitivity suite that includes both within-surface summary variations and one outside-surface estimator perturbation.

15.3. Results

Reachability remains large and directionally stable across all three planned contrasts against `N3`, while Jacobian magnitude is weaker and more selective. Conventional spectral and complexity summaries also distinguish stages strongly, but they do not reconstruct the subject-level reachability contrasts especially well. The paper's most important reviewer-facing result is therefore bounded nonredundancy rather than additive classifier gain.

15.4. Discussion

The manuscript sharpens a key principle in the NDT line of work: a geometric summary does not become useful merely because it separates familiar conditions. It becomes useful if it survives skeptical checks, null controls, and comparison to simpler baselines. In that sense, the paper is less about "sleep proves consciousness theory" and more about how to stress-test a dynamical measurement in a regime where conventional EEG already works well.

16. Article 15: Reduced Dynamical Capacity Does Not Imply Unconsciousness

Sleep, Propofol, and Epilepsy Dissociate in Transport Capture and Pathological Deformation

16.1. Abstract

This comparative paper asks whether low local dynamical capacity can serve as a generic unconsciousness marker. Its answer is no: sleep, propofol sedation, and epilepsy can all show reduced capacity, but they dissociate once transport-innovation balance and pathological lock-in are measured explicitly. The manuscript therefore replaces a one-dimensional unconsciousness scalar with a multi-axis typology of reduced-capacity regimes.

16.2. Method

The study analyzes three datasets under a common framework: ANPHY sleep EEG, propofol EEG, and epilepsy iEEG. It decomposes the problem into three partially independent layers: capacity collapse, transport-innovation balance, and pathological lock-in. These layers are quantified using traversability, reachability, Jacobian, and block-Jacobian summaries, then tested both as descriptive contrasts and as incremental-validity candidates beyond classical spectral features.

16.3. Results

Capacity collapse appears across all three regime families, but the other layers dissociate them. Deep sleep shows low capacity while remaining relatively innovation-dominated; propofol produces a clearer capacity shift but weaker evidence for pathological capture at the analyzed depth; epilepsy shows the strongest joint signature of collapse, transport capture, and lock-in, together with substantial incremental validity beyond spectral baselines. The important result is therefore not "which state is most unconscious," but that different low-capacity regimes occupy different positions in geometric phenotype space.

16.4. Discussion

This paper is one of the clearest examples of mature claim discipline in the project. It explicitly rejects overinterpretation of any single summary and reframes low-capacity states as a family that must be dissociated mechanistically. That move substantially improves the clinical and conceptual precision of the broader NDT program.

17. Article 16: Sleep-Spindle-Rich Windows Are Associated With Local Meta-Noetic Phase-Space Shifts in N2 Sleep

An event-locked measurement-layer analysis of spindle-associated reconfiguration

17.1. Abstract

This paper moves from stage-level sleep contrasts to event-level analysis by asking whether spindle-rich N2 windows differ from matched non-spindle N2 windows in local state-space geometry. The manuscript treats spindles not as direct proof of memory writing, but as time-localized physiological events that may carry reproducible MNPS, stratified, reachability, or Jacobian signatures.

17.2. Method

Using ds005555 processed under the NeuralManifoldDynamics contract, the study performs event-locked comparisons around detector-defined N2 spindles, baseline-corrected subject summaries, matched-control windows, label permutations, time-shifted pseudo-events, and detector-source sensitivity analyses. The promoted endpoint is spindle-locked MNPS deviation, supplemented by Stratified 9D and MNJ summaries as well as a modest incremental-validity check against spectral power.

17.3. Results

The strongest and most reproducible spindle effect is a positive event-locked shift in MNPS deviation relative both to pre-event baseline and to matched non-spindle N2 controls. Stratified 9D results show that the change is not confined to one 3D composite and is strongest in deviation-family and entropy-memory-family coordinates. Jacobian changes are present but weaker, and geometry adds only modest incremental discrimination on top of already-strong spectral power. The retained interpretation is therefore a local measurement-layer reconfiguration claim rather than a full mechanistic theory of spindle function.

17.4. Discussion

The manuscript sharpens the spindle story in two useful ways. First, it relocates the question from broad sleep-stage differences to tightly matched within-stage events. Second, it demonstrates how bounded interpretation can be preserved even when an effect is strong: spindle windows look geometrically different, but the paper explicitly stops short of claiming that this alone proves memory consolidation or a specific structural-write mechanism.

18. Article 17: A Cytoskeletal-Dendritic Accessibility Model of Associative Memory

A branch-resolved bridge from biological memory theory to executable and open-data tests

18.1. Abstract

This paper develops a narrower biological memory theory in which associative recall depends not only on synaptic weights but also on a slower, branch-level accessibility structure distributed across dendrites and spines. The claim is deliberately restrained: structural accessibility is presented as a biologically motivated, executable, and empirically contactable hypothesis, not as a uniquely established molecular memory code.

18.2. Method

The manuscript works across three linked levels. First, it synthesizes active dendrites, spine compartmentalization, tagging/capture, local translation, replay, and cytoskeletal dynamics into a slow-accessibility hypothesis. Second, it instantiates that logic in a minimal branch-resolved simulator with replay-dependent structural writing and perturbation tests. Third, it confronts the model with several open datasets chosen as downstream signature tests rather than as direct microscopic validation.

18.3. Results

In simulation, branch-specific structural writing, replay-dependent consolidation, and damage/rescue asymmetries survive stronger comparator and ablation tests than simpler alternatives. In open data, the paper finds modest but structured signature support across three dataset families, including replay-adjacent enrichment, inter-plane dendritic coupling, and weaker cross-day stability under perturbation. The combined result is not proof of a hidden structural code, but evidence that the accessibility hypothesis survives initial executable and empirical sharpening.

18.4. Discussion

This is one of the most biologically ambitious papers in the corpus, but also one of the most careful about claim boundaries. Its contribution is to connect dendritic physiology, cytoskeletal plasticity, and memory linking to a formal and testable accessibility layer. In the larger NDT ecosystem, it also provides a plausible substrate-level complement to the more abstract geometric and dynamical articles.